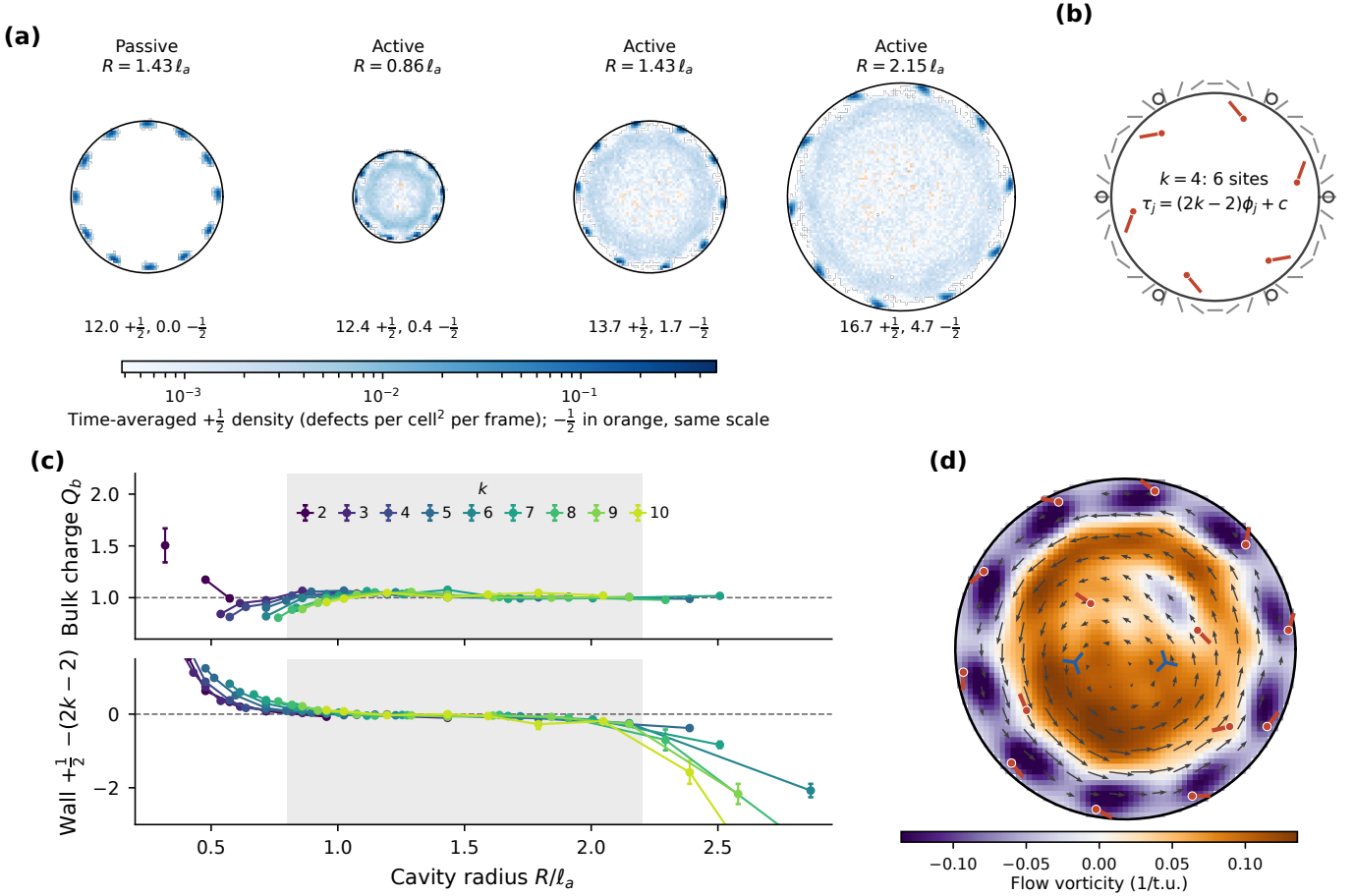

(a)
Passive
$R = 1.43\,\ell_a$
Active
$R = 0.86\,\ell_a$
Active
$R = 1.43\,\ell_a$
Active
$R = 2.15\,\ell_a$
12.0 $+\frac{1}{2}$, 0.0 $-\frac{1}{2}$
12.4 $+\frac{1}{2}$, 0.4 $-\frac{1}{2}$
13.7 $+\frac{1}{2}$, 1.7 $-\frac{1}{2}$
16.7 $+\frac{1}{2}$, 4.7 $-\frac{1}{2}$
10^{-3}
10^{-2}
10^{-1}
Time-averaged $+\frac{1}{2}$ density (defects per cell^2 per frame); $-\frac{1}{2}$ in orange, same scale
(b)
$k = 4$: 6 sites
$\tau_j = (2k-2)\phi_j + c$
(c)
Bulk charge Q_b
Wall $+\frac{1}{2}$ $-(2k-2)$
k
2 3 4 5 6 7 8 9 10
2.0
1.5
1.0
0
−2
0.5
1.0
1.5
2.0
2.5
Cavity radius R/ℓ_a
(d)
−0.10 −0.05 0.00 0.05 0.10
Flow vorticity (1/t.u.)